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FLUID HEATING DEVICE FOR VEHICLE

Abstract

A fluid heating device for a vehicle includes: a tank that includes a flow path that directs a fluid; a heating unit that heats the fluid by heating the tank; and a case that houses the tank and the heating unit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims priority of Japanese Patent Application No.

FIELD

[0002] The present disclosure relates to a fluid heating device for a vehicle.

BACKGROUND

[0003] Patent Literature (PTL) 1 discloses a fluid heating device that heats a fluid. The fluid heating device includes: a heater unit that includes a heater that generates heat and a heating unit that is heated by the heater; and a case that houses the heating unit and in which a fluid for heat exchange with the heating unit flows.

CITATION LIST

Patent Literature

[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2017-89915

SUMMARY

[0005] However, the fluid heating device according to PTL 1 can be improved upon.

[0006] In view of this, the present disclosure provides a fluid heating device for a vehicle capable of improving upon the above related art.

[0007] A fluid heating device for a vehicle according to one aspect of the present disclosure includes: a tank that includes a flow path that directs a fluid; a heating unit that heats the fluid by heating the tank; and a case that houses the tank and the heating unit.

[0008] A fluid heating device for a vehicle according to the present disclosure is capable of improving upon the above related art.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0010] FIG. 1 is a perspective view of a fluid heating device for a vehicle according to an embodiment.

[0011] FIG. 2 is an exploded perspective view of the fluid heating device for the vehicle according to the embodiment.

[0012] FIG. 3 is a plan view illustrating a state in which a tank is inserted to a case.

[0013] FIG. 4 is a lateral view of the fluid heating device for the vehicle according to the embodiment.

[0014] FIG. 5 is a cross-sectional view of the fluid heating device for the vehicle taken along line V-V in FIG. 1.

DESCRIPTION OF EMBODIMENT

[0015] Hereinafter, an embodiment will be specifically described with reference to the drawings.

[0016] It should be noted that the embodiment described below shows a general or specific example. The numerical values, shapes, materials, structural elements, arrangement positions and connection forms of the structural elements, and the like shown in the following embodiment are examples, and are not intended to limit the present disclosure. Moreover, among the structural elements in the embodiment below, the structural elements not described in independent claims will be described as arbitrary structural elements.

[0017] Note that the respective figures are schematic diagrams and are not necessarily precise illustrations. Additionally, the structural elements that are essentially the same share like reference numerals in the respective figures.

[0018] Moreover, in the embodiment described below, in FIG. 1, a positive X-axis direction is defined as the direction in which a tank is inserted into a first opening, a positive Y-axis direction is

defined as the direction of a circuit board relative to the cover, and a positive Z-axis direction is defined as the direction perpendicular to the positive X-axis direction and the positive Y-axis direction, and the direction of a second housing relative to a first housing. The directions in FIG. 1 may be applied to FIG. 2 and subsequent figures.

[0019] Moreover, in the embodiment described below, the expressions such as “X-axis direction” and “elongated” are used. For example, “X-axis direction” not only means being perfectly X-axis direction, and “elongated” not only means being perfectly elongated, but also mean being substantially X-axis direction and being substantially elongated, that is, mean a state including an error of, for example, approximately several percent. In addition, “X-axis direction” and “elongated” respectively mean being X-axis direction and elongated to the extent that the effects of the present disclosure can be achieved. The same applies to expressions using other direction and shape.

Embodiment

[0020] First, a configuration of fluid heating device **1** for a vehicle will be described with reference to FIG. 1 to FIG. 5.

[0021] FIG. 1 is a perspective view of fluid heating device **1** for a vehicle according to an embodiment. FIG. 2 is an exploded perspective view of fluid heating device **1** for the vehicle according to the embodiment. FIG. 3 is a plan view illustrating a state in which tank **10** is inserted into case **40**. FIG. 4 is a lateral view of fluid heating device **1** for the vehicle according to the embodiment. FIG. 5 is a cross-sectional view of fluid heating device **1** for the vehicle taken along line V-V in FIG. 1.

[0022] As illustrated in FIG. 1, fluid heating device **1** for the vehicle is applied to vehicle air conditioning devices, vehicle batteries, and the like mounted in vehicles, such as EVs (Electric Vehicles), HEVs (Hybrid Electric Vehicles), and PHEVs (Plug-in Hybrid Electric Vehicles). When fluid heating device **1** for the vehicle is applied to a vehicle air conditioning device, for example, fluid heating device **1** for the vehicle heats the fluid when the vehicle air conditioning device performs a heating operation. This allows the vehicle air conditioning device to deliver warm air into the vehicle cabin.

[0023] As illustrated in FIG. 1 and FIG. 2, fluid heating device **1** for the vehicle includes tank **10**, heating unit **20**, circuit board **30**, case **40**, and cover **50**.

[0024] Tank **10** is slim and elongated. Specifically, tank **10** is a container that is slim in the Z-axis direction and elongated in the X-axis direction. In other words, tank **10** is a container that is in a shape of a flat plate that is parallel to the XY plane.

[0025] Tank **10** may include a flow path therein for directing the fluid. Specifically, tank **10** includes supply port **11** through which the fluid is supplied from the outside into tank **10** and discharge port **12** through which the fluid that has passed through the flow path of tank **10** is discharged to the outside. Supply port **11** and discharge port **12** are aligned on lid **13** of tank **10** along the Y-axis direction. Each of supply port **11** and discharge port **12** is a connection port that can be connected to a hose (not illustrated). Hoses are connected to supply port **11** and discharge port **12**, allowing the fluid to flow in and out of tank **10**. The fluid is, for example, a liquid such as a coolant.

[0026] Lid **13** is disposed at the end portion of tank **10** in the negative X-axis direction, and is integrated with tank **10**. The term “integrated” means that lid **13** and tank **10** cannot be separated from each other without destroying lid **13** and tank **10**.

[0027] Heating unit **20** is disposed in tank **10**. Heating unit **20** is a heater, and is capable of heating tank **10**. With this, the fluid supplied through supply port **11** of tank **10** is heated by heating unit **20** as the fluid passes through tank **10**, and the heated fluid can be discharged through discharge port **12**.

[0028] Circuit board **30** is in a shape of a plate elongated along the X-axis direction, and is held in case **40** so as to be parallel to the XZ plane.

[0029] Circuit board **30** is capable of driving and controlling heating unit **20**. Circuit board **30** is capable of adjusting the temperature, heating period and the like of the fluid in tank **10** by driving and controlling heating unit **20**.

[0030] Case **40** houses tank **10**, heating unit **20**, and circuit board **30**. Case **40** is elongated in the X-axis direction, and is L-shaped when viewed along the X-axis direction. Case **40** is, for example, made of a metal material.

[0031] Specifically, case **40** includes: elongated first housing **110** that houses elongated tank **10**; and second housing **120** that is elongated along the longitudinal direction of first housing **110**, connected to first housing **110**, and extends in the direction intersecting first housing **110**.

[0032] First housing **110** is slim in the Z-axis direction and elongated in the X-axis direction so as to house tank **10**. In other words, first housing **110** is a container that is parallel to the XY plane.

[0033] First housing **110** includes first opening **112** at a first longitudinal end portion of first housing **110**. First opening **112** is open to allow tank **10** to be inserted. First opening **112** is positioned on the negative X-axis side of first housing **110**. This allows tank **10** to be inserted into first opening **112** of first housing **110** along the positive X-axis direction.

[0034] First housing **110** includes first bottom portion **111** at a second longitudinal end portion of first housing **110**. First bottom portion **111** serves as a bottom wall of first housing **110** on the positive X-axis side when tank **10** is inserted into first opening **112** of first housing **110**.

[0035] As illustrated in (a) and (b) of FIG. 3, first guide portion **117** is disposed on first bottom portion **111**. Tank **10** includes second guide portion **17** at front end portion **15** in the direction of insertion into first opening **112**. Front end portion **15** of tank **10** is the end portion of tank **10** on the positive X-axis side. First guide portion **117** includes inclined surface **117a** that is inclined upward toward the positive X-axis direction. Second guide portion **17** includes inclined surface **17a** that is inclined downward toward the negative X-axis direction. Therefore, when tank **10** is inserted into first opening **112**, second guide portion **17** on front end portion **15** of tank **10** is engaged with first guide portion **117** of first bottom portion **111**, so that second guide portion **17** abuts the surface of first guide portion **117** and positioning is achieved. In such a manner, first guide portion **117** is capable of guiding second guide portion **17** of tank **10** inserted through first opening **112**.

[0036] First bottom portion **111** includes first fastened portion **116** for fastening to tank **10**. As illustrated in FIG. 2 and FIG. 3, tank **10** includes first fastening portion **16** at front end portion **15** of tank **10** in the direction of insertion into first opening **112**. When tank **10** is inserted into first opening **112** and second guide portion **17** is guided to first guide portion **117**, positioning of first fastening portion **16** and first fastened portion **116** are achieved so that first fastening portion **16** and first fastened portion **116** can be fastened.

[0037] In the present embodiment, first fastening portion **16** and first fastened portion **116** are fastened by a screw, bolt, or the like. Each of first fastening portion **16** and first fastened portion **116** may be, for example, a claw-shaped fastening structure. Moreover, one of first fastening portion **16** or first fastened portion **116** may be a claw-shaped fastening structure, and the other of first fastening portion **16** or first fastened portion **116** may be a hole for hooking the claw. Other known configurations may be used as long as first fastening portion **16** and first fastened portion **116** can be fastened.

[0038] Heating unit **20** is disposed in tank **10**. When tank **10** is housed in first housing **110**, heat insulation layer K is provided between heating unit **20** and the inner wall surface of first housing **110**. Heat insulation layer K is, for example, a heat insulation material or an air layer provided between heating unit **20** and the inner wall surface of first housing **110**. When heat insulation layer K is a heat insulation material, a heat insulation material may be pasted to the inner wall surface of heating unit **20** or first housing **110**. In the present embodiment, heat insulation layer K is an air layer. Therefore, heating unit **20** and first housing **110** are spaced apart from each other to avoid contact between heating unit **20** and first housing **110**.

[0039] As illustrated in FIG. 2, first housing **110** is elongated along the longitudinal direction of

tank **10**, and includes first wall portion **113** that is connected to second housing **120**. First wall portion **113** of first housing **110** is positioned on the positive Z-axis side of first housing **110** so as to be parallel to the XY plane.

[0040] Second housing **120** is elongated along the longitudinal direction of tank **10**, and includes second wall portion **123** that is raised from first wall portion **113**. As illustrated in FIG. 4, when viewed along the X-axis direction, first wall portion **113** in FIG. 1 and second wall portion **123** are L-shaped. Connectors **31** for electrically connecting to circuit board **30** are attached to second wall portion **123**. In the present embodiment, power supply connector **31a** for supplying power to circuit board **30** and communication connector **31b** for performing communication with circuit board **30** are aligned on second wall portion **123** in the X-axis direction. Power supply connector **31a** and communication connector **31b** are examples of connectors **31**.

[0041] When case **40** is viewed along the longitudinal direction of case **40** and the outline of case **40** is defined by a rectangle (indicated by a double-dotted line in FIG. 4), connectors **31** are disposed on second housing **120**, at a position within the imaginary rectangular frame indicated by the double-dotted line and adjacent to first housing **110**. Connectors **31** protrude from second wall portion **123** in the positive Y-axis direction. The height of each connector **31** is lower than the height of first wall portion **113** in the positive Y-axis direction. Therefore, connectors **31** are positioned within the imaginary rectangular frame indicated by the two dotted lines.

[0042] Tank **10** inserted through first opening **112** includes lid **13** that is positioned to cover first opening **112**. Lid **13** includes second fastening portion **14** that is fastened to first opening **112**. First opening **112** includes second fastened portion **114** that is fastened to second fastening portion **14**.

[0043] Lid **13** is a plate-shaped member positioned at the end portion of tank **10** on the negative X-axis side. Lid **13** is held on tank **10** at an orientation that is parallel with the YZ plane. In the present embodiment, lid **13** includes a plurality of second fastening portions **14**. Second fastening portions **14** are disposed on the edge of lid **13**. Therefore, second fastened portions **114** are arranged at first opening **112** on a one-to-one correspondence with second fastening portions **14**.

[0044] In the present embodiment, second fastening portions **14** and second fastened portions **114** are fastening holes that are fastened by screws, bolts, or the like. Each of second fastening portion **14** and second fastened portion **114** may be, for example, a claw-shaped fastening structure.

Moreover, one of second fastening portion **14** or second fastened portion **114** may be a claw-shaped fastening structure, and the other of second fastening portion **114** or second fastened portion **14** may be a hole for hooking the claw. Other known configurations may be used as long as second fastening portions **14** and second fastened portions **114** can be fastened.

[0045] In such a manner, first fastening portion **16** and first fastened portion **116** on the positive X-axis side of tank **10** are fastened, and second fastening portion **14** and second fastened portion **114** on the negative X-axis side of tank **10** are fastened. Therefore, tank **10** can be firmly supported by case **40** against not only the collision of an object with the vehicle but also against the vibrations of vehicle driving.

[0046] Supply port **11** and discharge port **12** are provided on surface **13a** of lid **13**. Supply port **11**, discharge port **12**, and second fastening portions **14** are arranged closely. In other words, second fastening portions **14** are arranged to surround supply port **11** and discharge port **12**. Hoses are connected to supply port **11** and discharge port **12**, so that the hoses are disposed to extend from lid **13** in the negative X-axis direction. This makes the components included in the vehicle unlikely to come into contact with second fastening portions **14** when the vehicle collides with an object.

[0047] Case **40** includes second opening **122** that is open across first housing **110** and second housing **120**, on the face of case **40** on the negative Y-axis side. Second opening **122** is an insertion hole for housing circuit board **30** in case **40**. Circuit board **30** housed in case **40** is disposed between the plane defined by second opening **122** and tank **10** and across first housing **110** and second housing **120**. In other words, as illustrated in FIG. 5, circuit board **30** is positioned across first housing space **S1** of first housing **110** and second housing space **S2** of second housing **120**.

Circuit board **30** is held in case **40** so as to be parallel to the plane defined by second opening **122**, i.e., parallel to the XZ plane and intersect the imaginary XY plane along slim tank **10**.

[0048] As illustrated in FIG. 2 and FIG. 5, cover **50** is capable of covering second opening **122** of case **40**. Specifically, even though circuit board **30**, heating unit **20**, and tank **10** are housed in case **40**, cover **50** is capable of covering second opening **122** of case **40**. Cover **50** is held at second opening **122** of case **40** at an orientation that is parallel to the XZ plane.

[0049] For example, as illustrated in FIG. 2, cover **50** includes a plurality of third fastening portions **54**. Third fastening portions **54** are disposed on the edge of cover **50**. Therefore, third fastened portions **124** are disposed at second opening **122** in a one-to-one correspondence with third fastening portions **54**.

[0050] In the present embodiment, third fastening portion **54** and third fastened portion **124** are fastening holes fastened by screws, bolts, or the like. Each of third fastening portion **54** and third fastened portion **124** may be, for example, a claw-shaped fastening structure. Moreover, one of third fastening portion **54** or third fastened portion **124** may be a claw-shaped fastening structure, and the other of third fastened portion **124** or third fastening portion **54** may be a hole for hooking the claw. Other known configurations may be adopted as long as third fastening portion **54** and third fastened portion **124** can be fastened.

[0051] Fluid heating device **1** for the vehicle may further include thermal interface material **33**.

[0052] For example, circuit board **30** includes thermal interface material **33** that connects circuit board **30** and case **40**. Thermal interface material **33** is capable of transferring the heat from heated circuit board **30** to case **40**. In the present embodiment, thermal interface material **33** is disposed between electronic component **30a** on the positive Y-axis side of circuit board **30** and second wall portion **123** of second housing **120**. Thermal interface material **33** is applied to the surface of electronic component **30a** on the positive Y-axis side. Thermal interface material **33** is, for example, heat dissipating silicone.

[0053] Fluid heating device **1** for the vehicle may further include detector **32**.

[0054] For example, circuit board **30** may include detector **32** for detecting leakage of the fluid in case **40**. To enable detection of the liquid leakage from tank **10** in case **40**, detector **32** is disposed at the end portion of circuit board **30** on the negative Z-axis side, so that a pair of terminals is positioned vertically below circuit board **30**.

[0055] Circuit board **30** is capable of stopping supplying power to heating unit **20**, when detector **32** detects the leakage of the fluid.

[0056] Depending on the orientation of fluid heating device **1** for the vehicle when mounted in the vehicle, detector **32** may be disposed at another position on circuit board **30**. In other words, detector **32** may be disposed on circuit board **30** such that the fluid leakage can be detected before the fluid reaches circuit board **30** when the fluid leakage occurs.

[0057] Fluid heating device **1** for the vehicle may be disposed in the vehicle such that first housing **110** of case **40** is positioned on the front side of the vehicle (e.g., front bumper side) and second housing **120** of case **40** is positioned on the rear side of the vehicle. In this case, when the front of the vehicle collides with an object, and even if first housing **110** of fluid heating device **1** for the vehicle is damaged by the impact, since circuit board **30** is positioned on the rear side of first housing **110**, it is expected to prevent circuit board **30**, to which high voltage power is supplied, from being damaged. As for connectors **31**, since first housing **110**, which protrudes further in the positive Y-axis direction relative to connectors **31**, is likely to collide with equipment inside the vehicle first, it is expected to prevent connectors **31**, to which high-voltage power is supplied, from being damaged. In this case, since supply port **11** and discharge port **12** are disposed on a lateral side of fluid heating device **1** for the vehicle, it is also expected to prevent supply port **11** and discharge port **12** from being damaged.

<Operational Effects>

[0058] Next, the operational effects of fluid heating device **1** for the vehicle according to the

present embodiment will be described.

[0059] When a vehicle collides with an object, the case of the fluid heating device for the vehicle mounted in the vehicle may be damaged. In this case, the fluid inside the case may leak, affecting the devices inside the vehicle.

[0060] In view of the above, as described above, fluid heating device **1** for a vehicle according to technique **1** of the present embodiment, includes: tank **10** that includes a flow path for directing a fluid; heating unit **20** that heats the fluid by heating the tank **10**; and case **40** that houses tank **10** and heating unit **20**.

[0061] With this, tank **10** is better protected because case **40** covers the outer peripheral surface of tank **10** in which the fluid is stored. Therefore, it is expected to protect tank **10** even when case **40** is damaged by a collision between the vehicle and an object.

[0062] Therefore, according to the present disclosure, it is possible to prevent the fluid from leaking even when the vehicle collides with an object.

[0063] In particular, case **40** is also capable of protecting heating unit **20**, which is a high-voltage component, because heating unit **20** is housed in case **40**. This is expected to improve the safety in fluid heating device **1** for the vehicle.

[0064] Moreover, fluid heating device **1** for the vehicle according to technique **2** of the present embodiment is fluid heating device **1** for the vehicle according to technique **1**. In this case, heating unit **20** is disposed in tank **10**.

[0065] This allows heating unit **20** to heat tank **10**, thus warming the fluid directed inside tank **10**. As a result, a vehicle air conditioning device is capable of using the heated fluid to deliver warm air into the vehicle cabin when performing a heating operation.

[0066] Moreover, fluid heating device **1** for the vehicle according to technique **3** of the present embodiment is fluid heating device **1** for the vehicle according to technique **1** or **2**. In this case, tank **10** is elongated, case **40** includes first housing **110** that houses tank **10**, first housing **110** being elongated, first housing **110** includes first opening **112** at a first longitudinal end portion of first housing **110**, and, first opening **112** is open to allow tank **10** to be inserted.

[0067] Tank **10** and case **40** according to the present embodiment are elongated in the X-axis direction. However, for example, when the first opening is provided along the longitudinal direction of the case, the number of fastening portions and fastened portions for fastening the tank and the case would increase.

[0068] However, in the present embodiment, tank **10** is inserted from a lateral face of case **40**, i.e., first opening **112** is provided on the lateral face of tank **10** (first longitudinal end portion of tank **10**). Therefore, it is difficult to increase the area of the opening of first opening **112**. As a result, it is possible to prevent the number of fastening portions and fastened portions (second fastening portions **14** and second fastened portions **114**) for connecting first opening **112** and tank **10** from increasing.

[0069] In addition, an increase in the number of fastening portions and fastened portions (second fastening portion **14** and second fastened portion **114**) can be reduced, so that an increase in the number of fastening members such as screws, bolts, or the like can also be reduced. As a result, it is possible to reduce an increase in the number of component, assembling steps, and weight in fluid heating device **1** for the vehicle.

[0070] Fluid heating device **1** for the vehicle according to technique **4** of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 3. In this case, case **40** includes: first housing **110** that is elongated in a first direction, extends in a second direction intersecting the first direction, and houses tank **10**; second housing **120** that is elongated along a longitudinal direction of first housing **110**, is connected to first housing **110**, and extends in a third direction intersecting the first direction and the second direction, fluid heating device **1** for the vehicle includes connector **31**, and when case **40** is viewed along a longitudinal direction of case **40**, and an outline of case **40** is defined by an imaginary rectangular frame, connector **31** is

disposed on second housing **120**, in a position within the imaginary rectangular frame and adjacent to first housing **110**.

[0071] This allows connectors **31** to be positioned at the corner portion formed by the L-shaped first and second housings **110** and **120** to prevent connectors **31** from protruding outward beyond the imaginary rectangular frame. This prevents the size of fluid heating device **1** for the vehicle from increasing.

[0072] In addition, connectors are usually made of resin members. The connectors are likely to be damaged when a vehicle collides with an object. The connectors are in a high voltage state to supply power to the heating unit. If the connectors are damaged, leakage of electricity from the connectors could occur. For this reason, the connectors should be protected by a strong case.

[0073] In view of the above, in the present embodiment, connectors **31** are positioned such that connectors **31** do not protrude beyond the imaginary rectangular frame of case **40**. By doing so, it is considered that case **40** will collide with the equipment inside the vehicle first, when the vehicle collides with an object. Therefore, connectors **31** are expected to be protected.

[0074] Moreover, fluid heating device **1** for the vehicle according to technique 5 of the present embodiment is fluid heating device **1** for the vehicle according to technique 4. In this case, first housing **110** includes first wall portion **113** that is elongated along a longitudinal direction of tank **10**, first wall portion **113** being connected to second housing **120**, second housing **120** includes second wall portion **123** that is elongated along the longitudinal direction of tank **10**, second wall portion **123** being raised from first wall portion **113**, and connector **31** is attached to second wall portion **123**.

[0075] This allows connectors **31** to be positioned on second wall portion **123**. Therefore, connectors **31** can be disposed within the imaginary rectangle and at the corner portion formed by the L-shaped first wall portion **113** and second wall portion **123**. This prevents the size of fluid heating device **1** for the vehicle from increasing.

[0076] In addition, since it is considered that case **40** will be damaged first when the vehicle collides with an object, connectors **31** are expected to be protected.

[0077] Moreover, fluid heating device **1** for the vehicle according to technique 6 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 3 to 5. In this case, first housing **110** includes: first opening **112** at a first longitudinal end portion of first housing **110**, and first bottom portion **111** at a second longitudinal end portion of first housing **110**. First bottom portion **111** includes first guide portion **117**, tank **10** includes second guide portion **17** at front end portion **15** of tank **10** in a direction in which tank **10** is inserted into first opening **112**, and first guide portion **117** guides second guide portion **17** of tank **10** inserted through first opening **112**.

[0078] With this, when tank **10** is inserted from first opening **112** of case **40** along the positive X-axis direction, first guide portion **117** on front end portion **15** of tank **10** is guided to second guide portion **17** on first bottom portion **111** of case **40**. The engagement of first guide portion **117** and second guide portion **17** allows front end portion **15** of tank **10** to be held at a predetermined orientation.

[0079] Moreover, fluid heating device **1** for the vehicle according to technique 7 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 3 to 6. In this case, first housing **110** includes: first opening **112** at a first longitudinal end portion of first housing **110**; and first bottom portion **111** at a second longitudinal end portion of first housing **110**. Tank **10** includes first fastening portion **16** at front end portion **15** of tank **10** in a direction in which tank **10** is inserted into first opening **112**. First bottom portion **111** includes first fastened portion **116** to be fastened to first fastening portion **16**.

[0080] With this, first guide portion **117** and second guide portion **17** are engaged to maintain front end portion **15** of tank **10** at a predetermined orientation such that first fastening portion **16** and first fastened portion **116** can be fastened. Therefore, front end portion **15** of tank **10** can be fixed to

first bottom portion **111** by fastening first fastening portion **16** and first fastened portion **116**.

[0081] In addition, when tank **10** is inserted into case **40**, second guide portion **17** contacts case **40**, thus preventing heating unit **20** from being damaged at the time of insertion of tank **10** into case **40**.

[0082] Moreover, fluid heating device **1** for the vehicle according to technique 8 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 3 to 7. In this case, fluid heating device **1** for the vehicle includes heat insulation layer K between heating unit **20** and an inner wall surface of first housing **110**.

[0083] With this, heat insulation layer K is provided between cover **50** and tank **10**, so that heated tank **10** can be thermally insulated by heat insulation layer K, making the heat from tank **10** less likely to be escaped into case **40**.

[0084] Moreover, fluid heating device **1** for the vehicle according to technique 9 of the present embodiment is fluid heating device **1** for the vehicle according to technique 8. In this case, heat insulation layer K is an air layer.

[0085] With this, an air layer is provided between cover **50** and tank **10**, so that heated tank **10** can be thermally insulated by the air layer, making the heat from tank **10** less likely to be escaped into case **40**. Moreover, by providing the air layer, it is possible to prevent the weight of fluid heating device **1** for the vehicle from increasing and the manufacturing cost of fluid heating device **1** for the vehicle from increasing due to an increase in the number of components.

[0086] Moreover, fluid heating device **1** for the vehicle according to technique 10 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 9. In this case, fluid heating device **1** for the vehicle further includes circuit board **30** that is housed in case **40** to drive and control heating unit **20**. Tank **10** is slim in a first direction and elongated, and circuit board **30** is held in case **40** at an orientation that intersects an imaginary plane (XY plane) that is perpendicular to the first direction.

[0087] This allows circuit board **30** to stand relative to slim tank **10**, thus ensuring the distance between circuit board **30** and tank **10**. This reduces the effects of heat from heated tank **10**.

[0088] The distance between circuit board **30** and tank **10** is secured. However, on the other hand, circuit board **30** and tank **10** are also in a distance relationship that allows direct connection between the connection terminals of circuit board **30** and the connection terminals of heating unit **20** in order for circuit board **30** to supply power to heating unit **20**. Therefore, in the present embodiment, circuit board **30** and tank **10** are arranged in such a manner that achieves both a reduction of the effects of heat from tank **10** to circuit board **30** and the electrical connection between circuit board **30** and heating unit **20**.

[0089] Since circuit board **30** is disposed in second housing **120**, circuit board **30** is disposed adjacent to second wall portion **123** of second housing **120**. Therefore, the heat emitted from circuit board **30** is more likely to be transferred to second wall portion **123**, and is more likely to be dissipated from circuit board **30**.

[0090] The heat emitted from circuit board **30** can also be transferred to the lateral sides of tank **10**.

[0091] Moreover, fluid heating device **1** for the vehicle according to technique 11 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 9. In this case, fluid heating device **1** further includes: circuit board **30** that is housed in case **40** to drive and control heating unit **20**. Case **40** includes: first housing **110** that is elongated in a first direction, extends in a second direction intersecting the first direction, and houses tank **10**; and second housing **120** that is elongated along a longitudinal direction of first housing **110**, is connected to first housing **110**, and extends in a third direction intersecting the first direction and the second direction, and circuit board **30** is disposed across first housing space S1 of first housing **110** and second housing space S2 of second housing **120**.

[0092] This allows circuit board **30** to be positioned across a space different from first housing space S1 (second housing space S2) in which tank **10** is disposed, while circuit board **30** is arranged to stand relative to slim tank **10**. Therefore, the distance between circuit board **30** and tank

10 can be secured. This reduces the effects of heat from heated tank **10**.

[0093] Moreover, since circuit board **30** is disposed in second housing **120**, circuit board **30** is disposed adjacent to second wall portion **123** of second housing **120**. Therefore, the heat emitted from circuit board **30** is transferred to second wall portion **123**, making the heat more likely to be dissipated from circuit board **30**.

[0094] Moreover, fluid heating device **1** for the vehicle according to technique 12 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 9. In this case, fluid heating device **1** for the vehicle further includes: circuit board **30** that is housed in case **40** to drive and control heating unit **20**. Case **40** includes: first housing **110** that is elongated in a first direction, extends in a second direction intersecting the first direction, and houses tank **10**; and second housing **120** that is elongated along a longitudinal direction of first housing **110**, is connected to first housing **110**, and extends in a third direction intersecting the first direction and the second direction, and case **40** includes second opening **122** to which circuit board **30** is inserted, second opening **122** being provided across first housing **110** and second housing **120**, and circuit board **30** is disposed between a plane defined by second opening **122** and tank **10**, and is held in case **40** at an orientation that is parallel to the plane defined by second opening **122**.

[0095] This allows fluid heating system **1** for the vehicle to be disposed in the vehicle such that first housing **110** is positioned on the front side of the vehicle and second housing **120** is positioned on the rear side of the vehicle. Even when the front side of the vehicle collides with an object, first housing **110** will hit the equipment inside the vehicle first. At this time, even when first housing **110** is damaged, it is expected that circuit board **30** positioned on the rear side of the vehicle will not be damaged.

[0096] Moreover, fluid heating device **1** for the vehicle according to technique 13 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 9. In this case, fluid heating device **1** for the vehicle further includes: circuit board **30** that is housed in case **40** to drive and control heating unit **20**; and thermal interface material **33** that connects circuit board **30** and case **40**.

[0097] With this, even when circuit board **30** generates heat by driving and controlling heating unit **20**, thermal interface material **33** is capable of releasing the heat to case **40**. This reduces an increase in the temperature of circuit board **30**.

[0098] Moreover, fluid heating device **1** for the vehicle according to technique 14 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 13. In this case, case **40** includes first housing **110** that is elongated and houses tank **10**; first housing **110** includes first opening **112** at a first longitudinal end portion of first housing **110**; tank **10** includes lid **13** that covers first opening **112** when tank **10** is inserted through first opening **110**. Lid **13** is integrated with tank **10**, and includes second fastening portion **14** that is fastened to first opening **112**, and first opening **112** includes second fastened portion **114** that is fastened to second fastening portion **14**.

[0099] This allows lid **13** integrated with tank **10** to be fastened to first opening **112** to seal first opening **112**, eliminating the need for a separate component to cover first opening **112**. As a result, an increase in the number of components, assembling steps, and weight of fluid heating system **1** for the vehicle can be reduced.

[0100] Furthermore, once first fastening portion **16** and first fastened portion **116** are fastened, front end portion **15** of tank **10** can be fixed to first bottom portion **111**, allowing tank **10** to be more firmly supported by case **40**. In other words, tank **10** can be firmly supported by case **40** against vehicle vibrations because the end portion of tank **10** on the positive X-axis side and the end portion of tank **10** on the negative X-axis side are fastened to case **40**.

[0101] Moreover, fluid heating device **1** for the vehicle according to technique 15 of the present embodiment is fluid heating device **1** for the vehicle according to technique 14. In this case, tank **10** includes: supply port **11** through which the fluid is supplied into tank **10** from outside tank **10**;

and discharge port **12** through which the fluid having passed through the flow path of tank **10** is discharged to the outside tank **10**, and supply port **11** and discharge port **12** are provided on surface **13a** of lid **13**.

[0102] This allows supply port **11** and discharge port **12** to be provided on the same surface **13a** of lid **13**, thus reducing an increase of the external size of tank **10** compared to when supply port **11** and discharge port **12** are provided on different surfaces.

[0103] Moreover, supply port **11** and discharge port **12** can be provided on the same surface **13a**, allowing the fluid supplied from supply port **11** to make a U-turn in tank **10** and to be discharged from discharge port **12**. This allows the flow path length from supply port **11** to discharge port **12** to be as long as possible. As a result, heating of the fluid in tank **10** by heating unit **20** is facilitated.

[0104] Moreover, fluid heating device **1** for the vehicle according to technique 16 of the present embodiment is fluid heating device **1** for the vehicle according to technique 15. In this case, supply port **11**, discharge port **12**, and second fastening portion **14** are closely arranged.

[0105] With this, hoses are connected to supply port **11** and discharge port **12**, so that the hoses extend from lid **13**. Since the hoses are disposed, no equipment is disposed around the hoses in the vehicle, and it is possible to prevent second fastening portion **14** from coming into contact with the equipment inside the vehicle. In addition, when the vehicle collides with an object, the equipment inside the vehicle is more likely to collide with the protruding hoses. This prevents second fastening portion **14** from colliding with the equipment inside the vehicle, thus preventing second fastening portion **14** and lid **13** from being damaged.

[0106] Moreover, fluid heating device **1** for the vehicle according to technique 17 of the present embodiment is fluid heating device **1** for the vehicle according to any one of techniques 1 to 16. In this case, fluid heating device **1** for the vehicle further includes: circuit board **30** that is housed in case **40** to drive and control heating unit **20**. Circuit board **30** includes detector **32** that detects a leakage of the fluid in case **40**, and circuit board **30** stops supplying power to heating unit **20** when detector **32** detects the leakage of the fluid.

[0107] With this, even when water leakage from tank **10** occurs, it is possible for detector **32** to detect the leakage. As a result, circuit board **30** is capable of stopping supplying power to heating unit **20** before circuit board **30** and heating unit **20** are exposed to water.

(Other Variations)

[0108] Although the fluid heating device for the vehicle according to the present disclosure has been described based on the embodiment, the present disclosure is not limited to the embodiment. Various modifications to the embodiment that can be conceived by those skilled in the art without departing from the teachings of the present disclosure are also included in the scope of the present disclosure.

[0109] In addition, a form obtained making various modifications conceivable by those skilled in the art to the embodiment described above, and a form realized by arbitrarily combining the structural elements and functions in the embodiment without departing from the teachings of the present disclosure are also included in the present disclosure.

[0110] While various embodiments have been described herein above, it is to be appreciated that various changes in form and detail may be made without departing from the spirit and scope of the present disclosure as presently or hereafter claimed.

FURTHER INFORMATION ABOUT TECHNICAL BACKGROUND TO THIS APPLICATION

[0111] The disclosure of the following patent application including specification, drawings, and claims are incorporated herein by reference in their entirety: Japanese Patent Application No. 2024-026181 filed on Feb. 26, 2024.

INDUSTRIAL APPLICABILITY

[0112] The present disclosure is usable for a fluid heating device for a vehicle, such as an air conditioning device mounted in the vehicle.

Claims

1. A fluid heating device for a vehicle, comprising: a tank that includes a flow path for directing a fluid; a heating unit that heats the fluid by heating the tank; and a case that houses the tank and the heating unit.
2. The fluid heating device for the vehicle according to claim 1, wherein the heating unit is disposed in the tank.
3. The fluid heating device for the vehicle according to claim 1, wherein the tank is elongated, the case includes a first housing that houses the tank, the first housing being elongated, the first housing includes a first opening at a first longitudinal end portion of the first housing, and the first opening is open to allow the tank to be inserted.
4. The fluid heating device for the vehicle according to claim 1, wherein the case includes: a first housing that is elongated in a first direction, extends in a second direction intersecting the first direction, and houses the tank; and a second housing that is elongated along a longitudinal direction of the first housing, is connected to the first housing, and extends in a third direction intersecting the first direction and the second direction, the fluid heating device for the vehicle comprises a connector, and when the case is viewed along a longitudinal direction of the case, and an outline of the case is defined by an imaginary rectangular frame, the connector is disposed on the second housing, in a position within the imaginary rectangular frame and adjacent to the first housing.
5. The fluid heating device for the vehicle according to claim 4, wherein the first housing includes a first wall portion that is elongated along a longitudinal direction of the tank, the first wall portion being connected to the second housing, the second housing includes a second wall portion that is elongated along the longitudinal direction of the tank, the second wall portion being raised from the first wall portion, and the connector is attached to the second wall portion.
6. The fluid heating device for the vehicle according to claim 3, wherein the first housing includes: a first opening at a first longitudinal end portion of the first housing; and a first bottom portion at a second longitudinal end portion of the first housing, the first bottom portion includes a first guide portion, the tank includes a second guide portion at a front end portion of the tank in a direction in which the tank is inserted into the first opening, and the first guide portion guides the second guide portion of the tank inserted through the first opening.
7. The fluid heating device for the vehicle according to claim 3, wherein the first housing includes: a first opening at a first longitudinal end portion of the first housing; and a first bottom portion at a second longitudinal end portion of the first housing, the tank includes a first fastening portion at a front end portion of the tank in a direction in which the tank is inserted into the first opening, and the first bottom portion includes a first fastened portion to be fastened to the first fastening portion.
8. The fluid heating device for the vehicle according to claim 3, further comprising: a heat insulation layer between the heating unit and an inner wall surface of the first housing.
9. The fluid heating device for the vehicle according to claim 8, wherein the heat insulation layer is an air layer.
10. The fluid heating device for the vehicle according to claim 1, further comprising: a circuit board that is housed in the case to drive and control the heating unit, wherein the tank is slim in a first direction and elongated, and the circuit board is held in the case at an orientation that intersects an imaginary plane that is perpendicular to the first direction.
11. The fluid heating device for the vehicle according to claim 1, further comprising: a circuit board that is housed in the case to drive and control the heating unit, wherein the case includes: a first housing that is elongated in a first direction, extends in a second direction intersecting the first direction, and houses the tank; and a second housing that is elongated along a longitudinal direction of the first housing, is connected to the first housing, and extends in a third direction intersecting the first direction and the second direction, and the circuit board is disposed across a first housing

space of the first housing and a second housing space of the second housing.

12. The fluid heating device for the vehicle according to claim 1, further comprising: a circuit board that is housed in the case to drive and control the heating unit, wherein the case includes: a first housing that is elongated in a first direction, extends in a section direction intersecting the first direction, and houses the tank; and a second housing that is elongated along a longitudinal direction of the first housing, is connected to the first housing, and extends in a third direction intersecting the first direction and the second direction, and the case includes a second opening to which the circuit board is inserted, the second opening being provided across the first housing and the second housing, and the circuit board is disposed between a plane defined by the second opening and the tank, and is held in the case at an orientation that is parallel to the plane defined by the second opening.

13. The fluid heating device for the vehicle according to claim 1, further comprising: a circuit board that is housed in the case to drive and control the heating unit; and a thermal interface material that connects the circuit board and the case.

14. The fluid heating device for the vehicle according to claim 1, wherein the case includes a first housing that is elongated, and houses the tank, the first housing includes a first opening at a first longitudinal end portion of the first housing, the tank includes a lid that covers the first opening when the tank is inserted through the first opening, the lid is integrated with the tank, and includes a second fastening portion that is fastened to the first opening, and the first opening includes a second fastened portion that is fastened to the second fastening portion.

15. The fluid heating device for the vehicle according to claim 14, wherein the tank includes: a supply port through which the fluid is supplied into the tank from outside the tank; and a discharge port through which the fluid having passed through the flow path of the tank is discharged to the outside the tank, and the supply port and the discharge port are provided on a surface of the lid.

16. The fluid heating device for the vehicle according to claim 15, wherein the supply port, the discharge port, and the second fastening portion are closely arranged.

17. The fluid heating device for the vehicle according to claim 1, further comprising: a circuit board that is housed in the case to drive and control the heating unit, wherein the circuit board includes a detector that detects a leakage of the fluid in the case, and the circuit board stops supplying power to the heating unit when the detector detects the leakage of the fluid.
